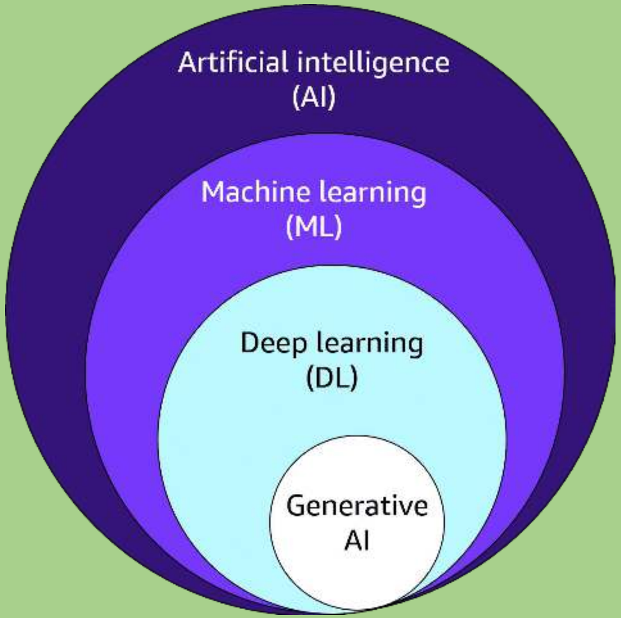

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Introduction to: AI / ML/ DL

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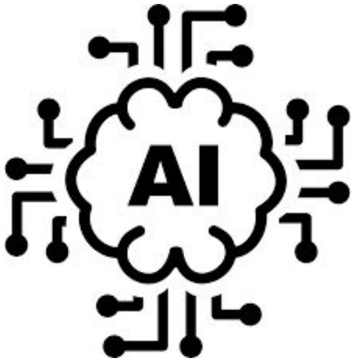
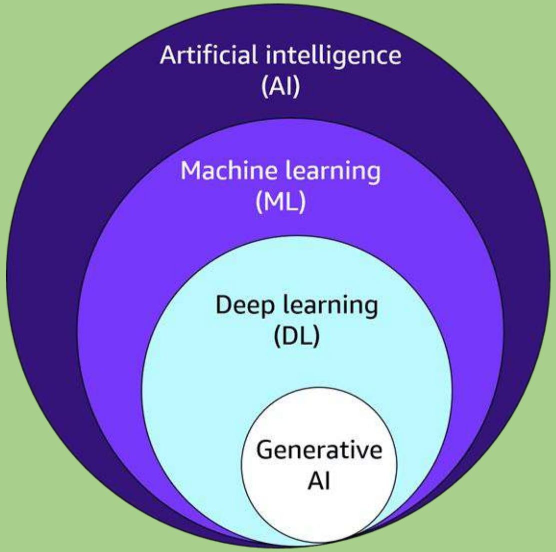
Artificial Intelligence

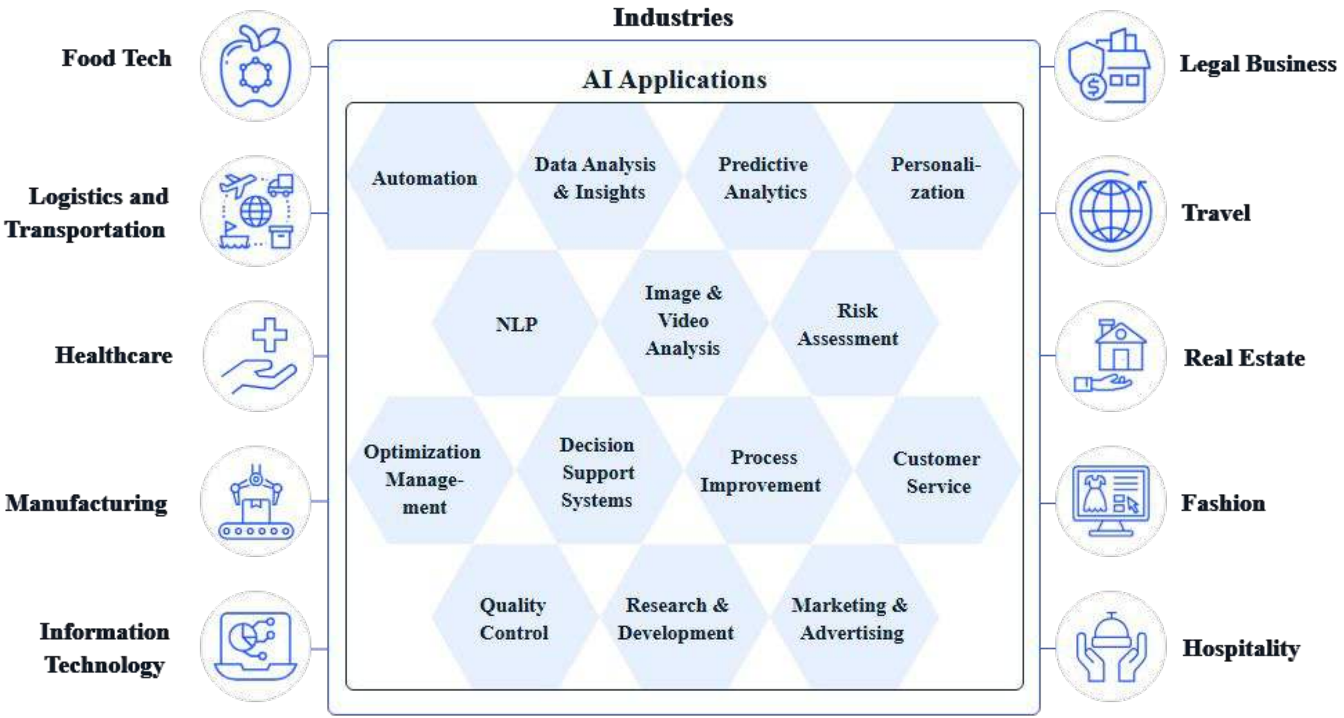
- **Definition**: AI refers to the broad field of computer science aimed at creating machines that can mimic human intelligence. It encompasses various technologies and techniques designed to enable machines to perform tasks that typically require human intelligence, such as reasoning, learning, decision-making, and problem-solving.
- **Scope**: AI is the **broadest category**, and ML, DL, and GenAI are subsets of AI. It includes everything from rule-based systems to more advanced approaches like machine learning and natural language processing.
- **Examples**:
  - Chatbots like Siri or Alexa
  - Self-driving cars
  - Chess-playing computers like Deep Blue
  - Rule-based expert systems (e.g., medical diagnosis systems)





Common Use Cases of AI

AI: Common Use Cases

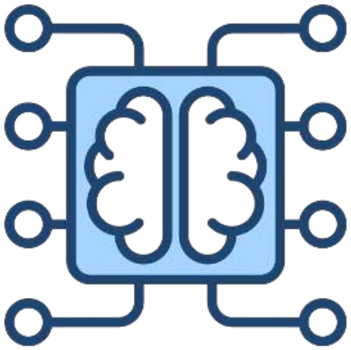


AI: When To Use ?

Scenario	Why Use AI	Example
Complex Tasks (Can't Code It)	When the problem requires mimicking human intelligence and decision-making	Chatbots, expert systems, self-driving cars
Scaling Expertise (Can't Scale It)	Automating intelligent tasks that require human-like reasoning at scale	Customer support automation, fraud detection
Adapting/Personalization	When systems need to improve and adapt based on environment and data	Personalized recommendations, adaptive learning
Autonomous/Unpredictable (Can't Track It)	When tasks require real-time, unpredictable decision-making	Autonomous vehicles, real-time analytics

Machine Learning

- **Definition**: ML is a **subset of AI** that focuses on creating systems that can learn from data, improve over time, and make decisions without being explicitly programmed for specific tasks. It uses algorithms to find patterns in data and make predictions or decisions based on those patterns.
- **How it Works**: In traditional programming, the programmer writes explicit rules, but in ML, the machine learns these rules by analyzing large datasets.
- **Examples**:
  - Recommendation engines (e.g., Netflix, Amazon recommendations)
  - Fraud detection systems in banking
  - Email spam filters
  - Predictive text on smartphones



What is ML Model?

A Machine Learning (ML) model is a mathematical representation or algorithm trained on data to recognize patterns, make predictions, or decisions without being explicitly programmed for specific tasks.

Key points:

- **Input**: The model takes data as input.
- **Training**: It learns patterns or relationships from this data during the training phase.
- **Output**: Once trained, it can predict outcomes or classify new, unseen data.
- **Examples**: Models can be linear regression, decision trees, neural networks, etc.
- **Purpose**: Automate decision-making or predictions based on learned data.

View only

Model = Algo (Data)

What is ML Model?

Model is the learnt Algo from Data

A Machine Learning (ML) model is a mathematical representation or algorithm trained on data to recognize patterns, make predictions, or decisions without being explicitly programmed for specific tasks.

Statistical Learning

S.L.

$y = 2.5 + 3.5x$

ML | DL | GenAI

Cannot be mathematically expressed as an eqn

Key points:

- **Input:** The model takes data as input.
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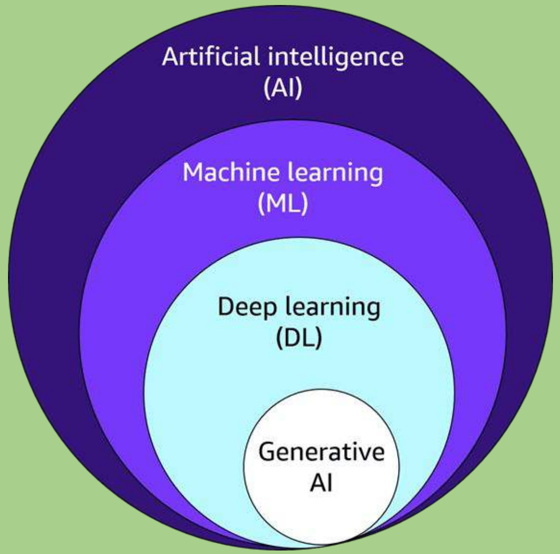
- **Output:** Once trained, it can predict outcomes or classify new, unseen data.
- **Examples:** Models can be linear regression, decision trees, neural networks, etc.
- **Purpose:** Automate decision-making or predictions based on learned data.

learn and find the Relationship | Pattern in the Data

Data → x, y



Machine Learning vs Traditional Programming

Traditional Programming

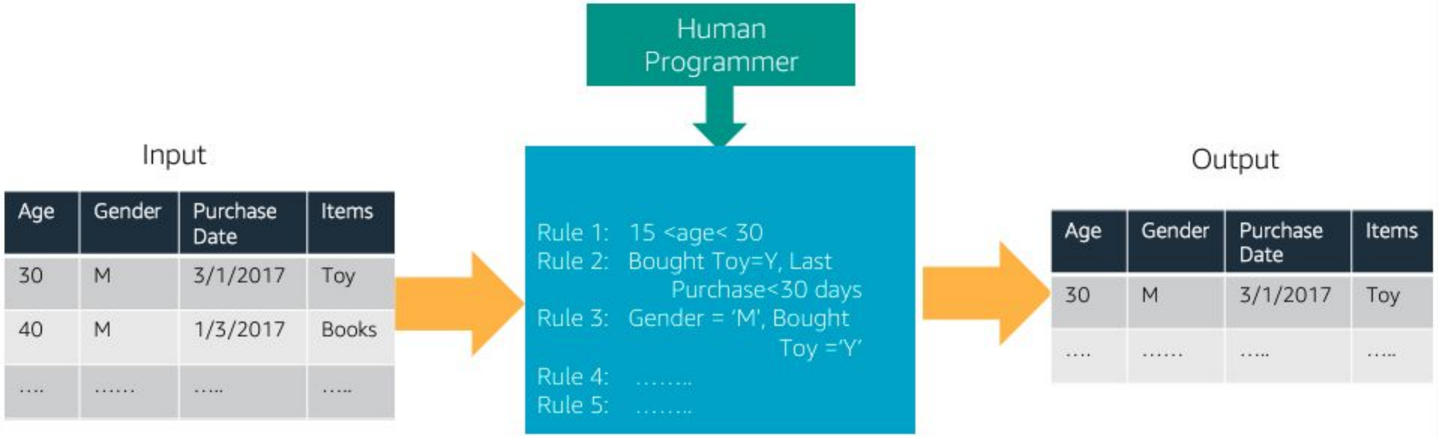
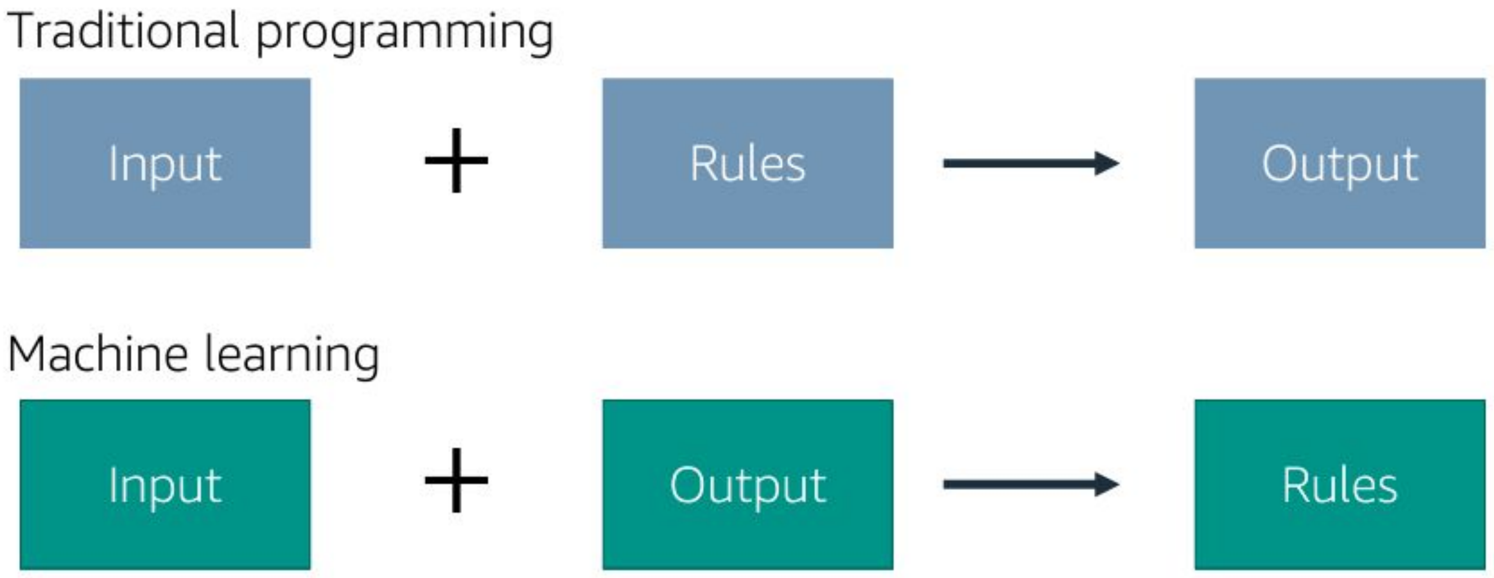
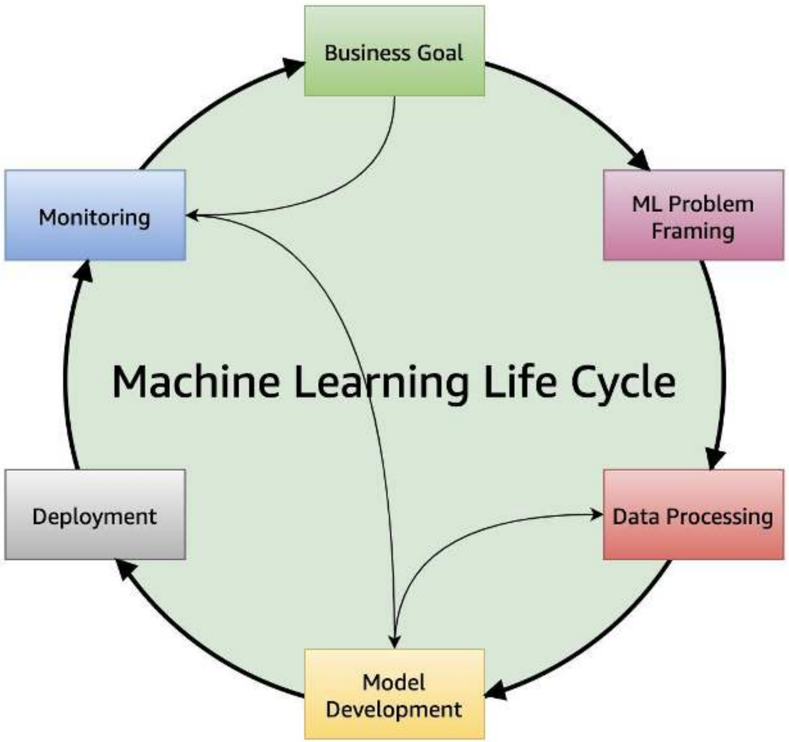


Image Source: amazon.com

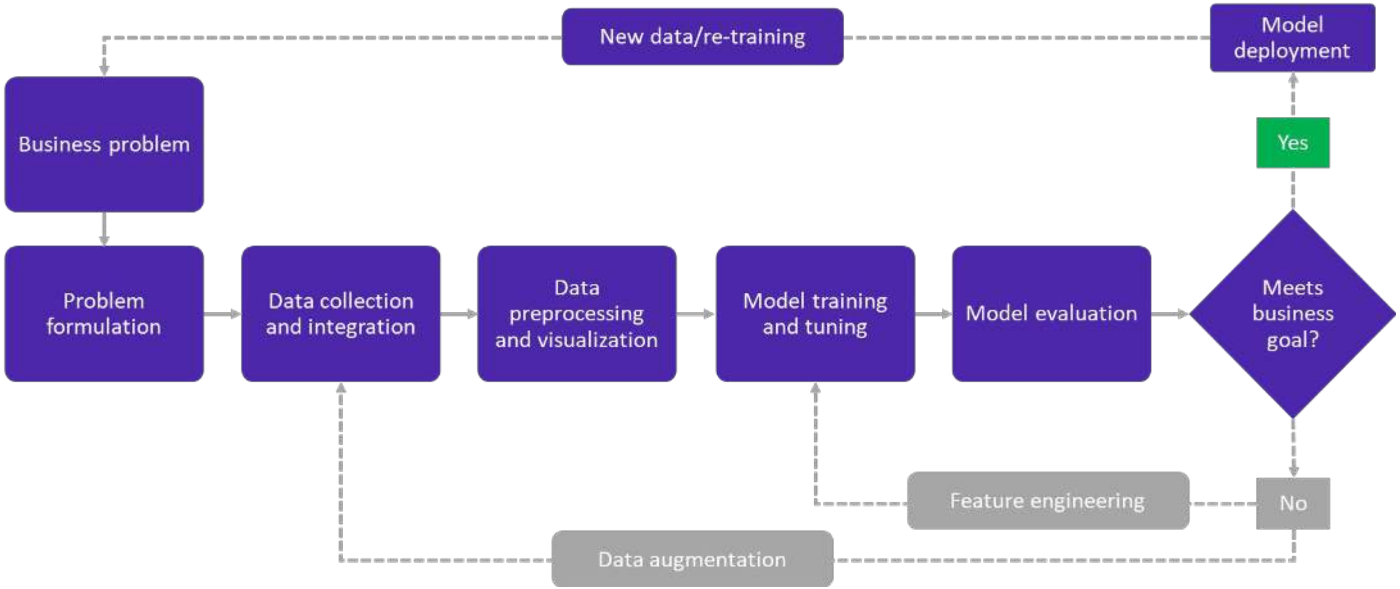
Traditional Programming vs ML



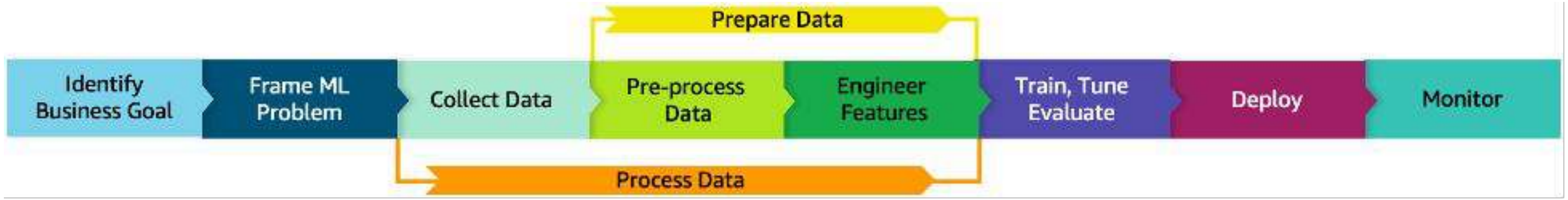
Machine Learning Lifecycle

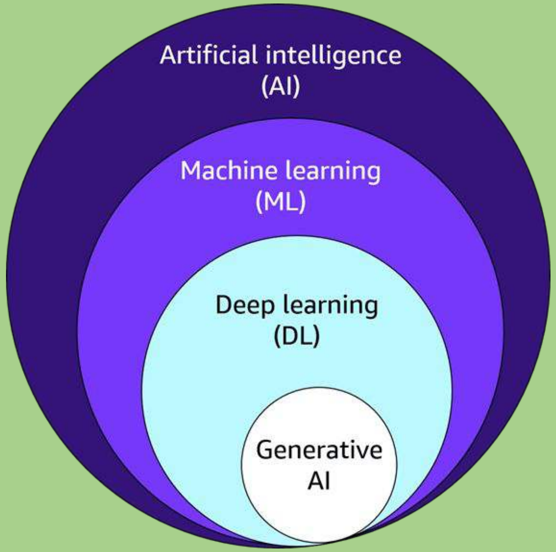


Traditional Machine Learning Pipeline

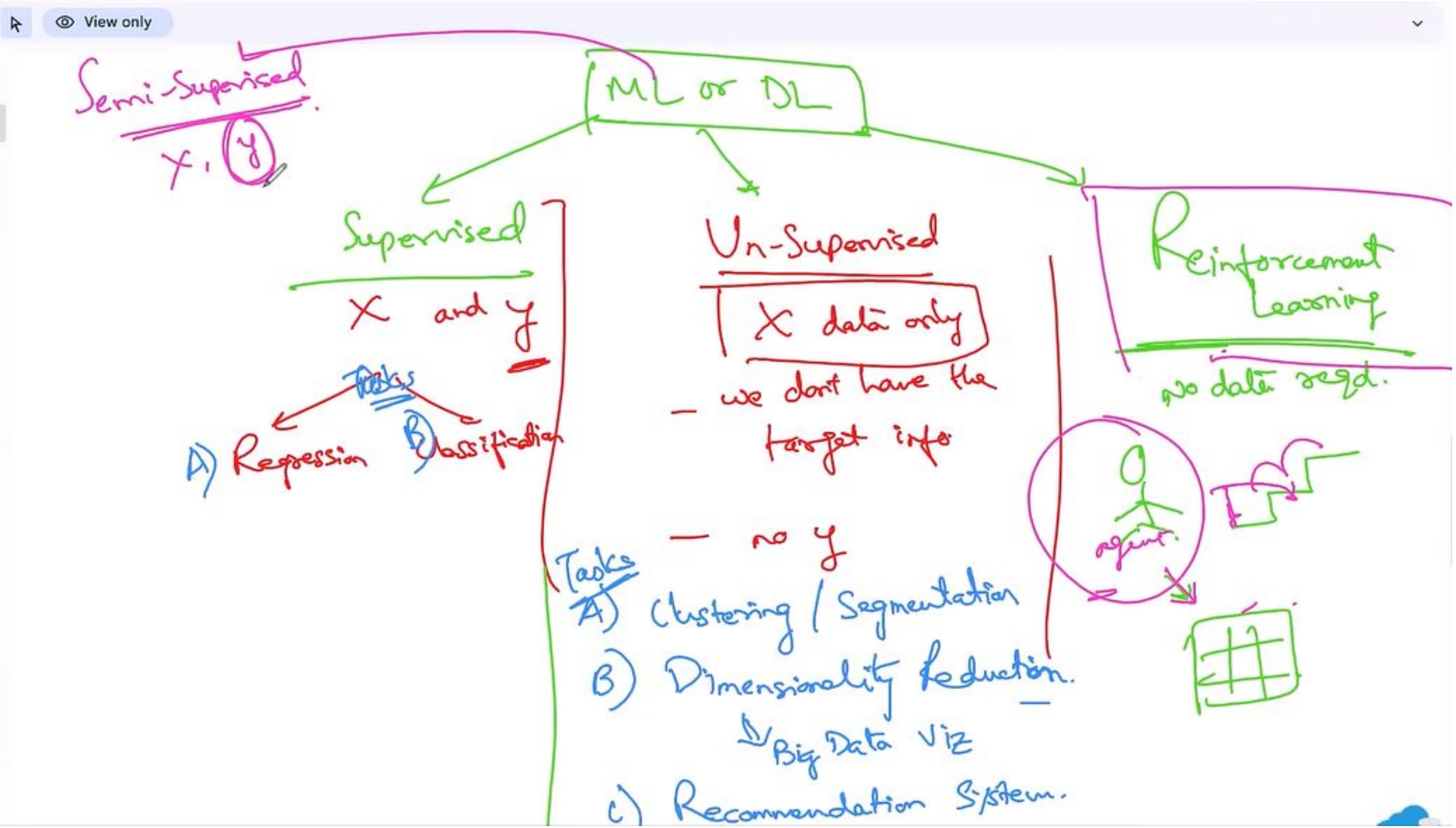


ML Lifecycle Phases

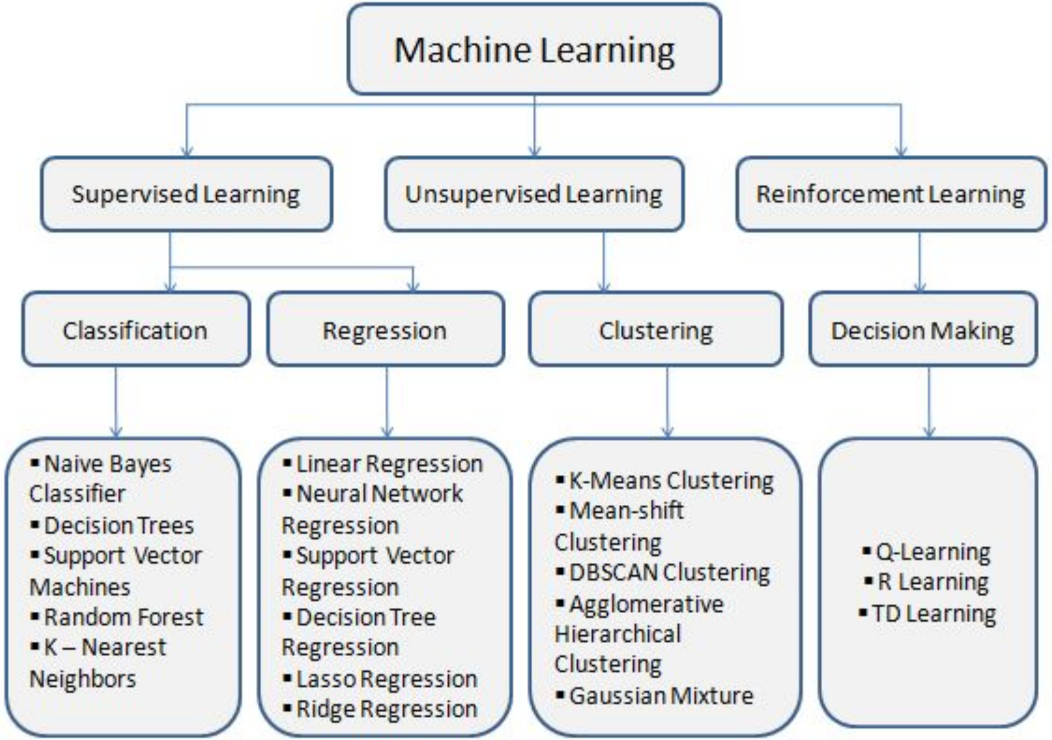




Machine Learning Types


View only
Semi-Supervised
X, y
ML or DL
Supervised
X and y
Tasks
A) Regression
B) Classification
Un-Supervised
X data only
- we don't have the target info
- no y
Tasks
A) Clustering / Segmentation
B) Dimensionality Reduction.
Big Data Viz
c) Recommendation System.
Reinforcement Learning
no data reqd.
agent

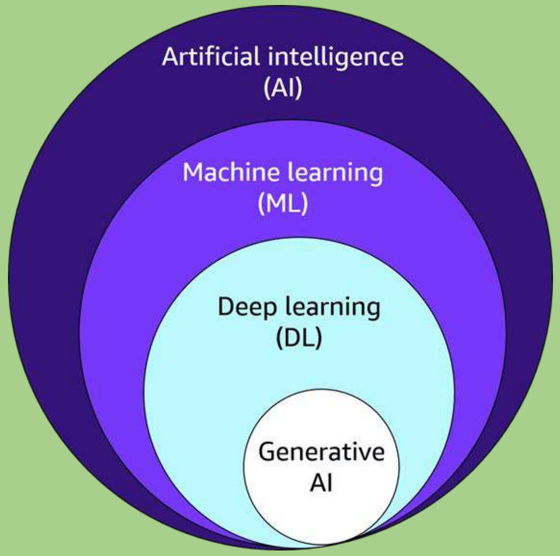
Machine Learning Types



Quiz Time

Which type of machine learning uses **reward and punishment** as part of its learning process?

A) Supervised Learning
B) Unsupervised Learning
C) Reinforcement Learning
D) Semi-supervised Learning



Common Use Cases of Machine Learning

ML: When To Use ?

Scenario	Why Use ML	Example
Complex Tasks (Can't Code It)	When the problem involves patterns and variability too complex for rules	Speech or image recognition, NLP
Scaling Expertise (Can't Scale It)	Automating human-like tasks that require expertise at scale	Recommendations, spam detection, fraud detection
Adapting/Personalization	When the system needs to learn and adapt over time based on new inputs	Personalized recommendations, adaptive learning
Autonomous/Unpredictable (Can't Track It)	When the task involves real-time, unpredictable decisions	Autonomous driving, real-time stock predictions

ML: Common Use Cases

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Financial Fraud Detection

- Interpretability & Explainability crucial for regulations & fairness
- Traditional ML models preferred because of their transparency with clear decision rules

Healthcare Diagnostics

- Accuracy & Interpretability critical
- Traditional ML on structured data:
    - Patient records
    - Lab results
    - Imaging data
- Reliable diagnoses & treatment recommendations
- Decision process auditable by medical experts

Autonomous Vehicles

- Safety & Reliability paramount
- Traditional:
    - Computer Vision
    - Object Detection
    - Rule-based Systems
- Preferred over Generative AI
- Robust & Predictable in real-time environments

Manufacturing Quality Control

- Traditional ML on sensor data, images, structured data
- Detect defects, anomalies, quality issues
- Precise and consistent results crucial for:
    - Quality control
    - Process optimization

Image Source: amazon.com

ML: Common Use Cases

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Credit Risk Assessment

- Transparency & Fairness critical for compliance
- Traditional models preferred:
    - Logistic Regression
    - Decision Trees
    - Credit Scoring
- Interpretable with clear decision criteria
- Avoid discrimination

Cybersecurity Threat Detection

- Traditional ML on network traffic, system logs, structured data
- Detect anomalies, intrusions, cyber threats
- High accuracy & reliability crucial:
    - Prevent false positives
    - Maintain system integrity

Predictive Maintenance

- Manufacturing, Aviation, Energy
- Sensor data, maintenance records
- Traditional models:
    - Time-series forecasting
    - Regression
- Accurate and interpretable
- Predict failures and maintenance needs

Image Source: amazon.com

Deep Learning

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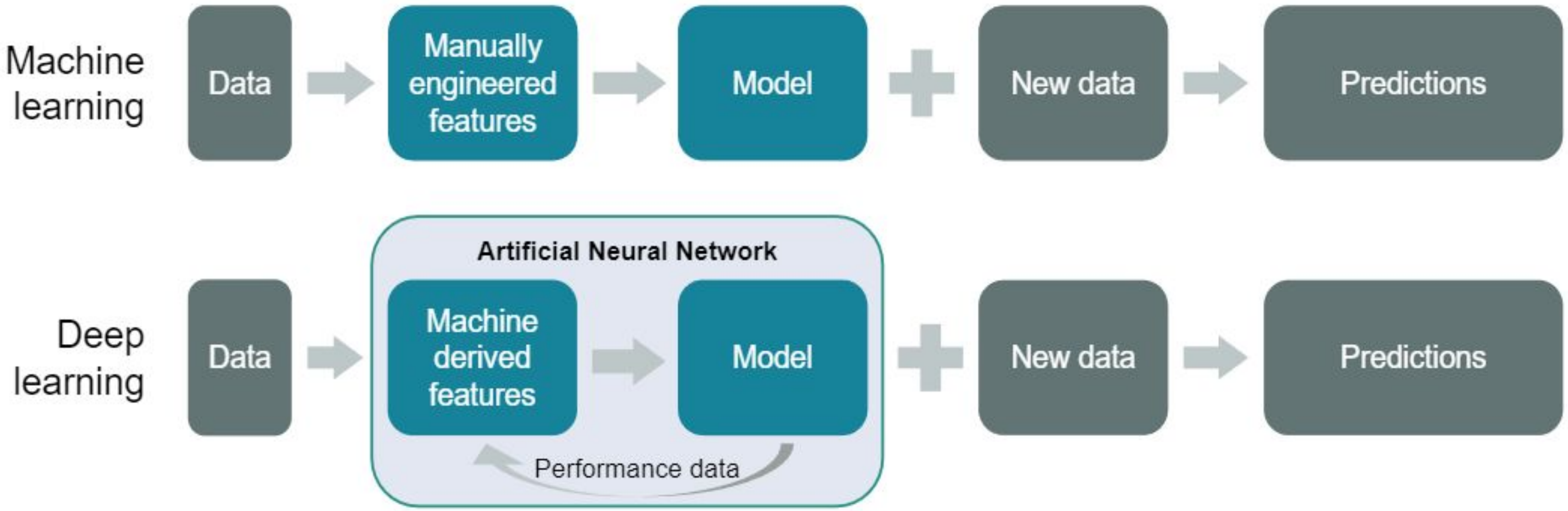
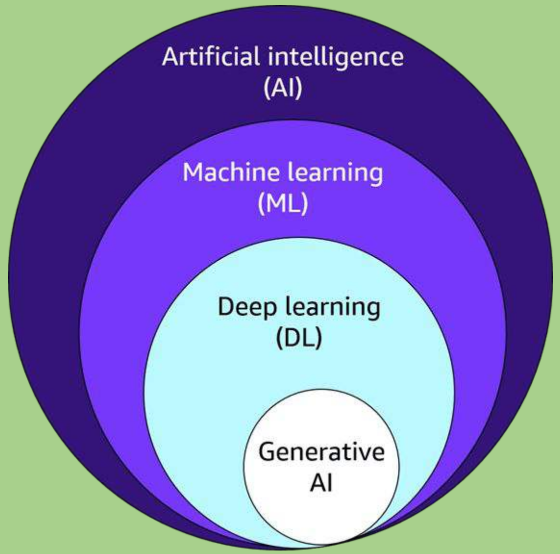
Deep Learning is a specialized branch of Machine Learning that uses neural networks with many layers (called **deep neural networks**) to model and understand complex patterns in data.

Key Points:

- **Structure:** Composed of multiple layers of interconnected nodes (neurons) that process data hierarchically.
- **Function:** Each layer extracts increasingly abstract features from the input.
- **Strength:** Excels at tasks like image recognition, natural language processing, speech recognition, and more.
- **Data Requirements:** Typically requires large amounts of data and computational power.
- **Examples:** Convolutional Neural Networks (CNNs) for images, Recurrent Neural Networks (RNNs) for sequences like text or speech.

How DL is different from ML?





Common Use Cases of DL

DL: When To Use ?

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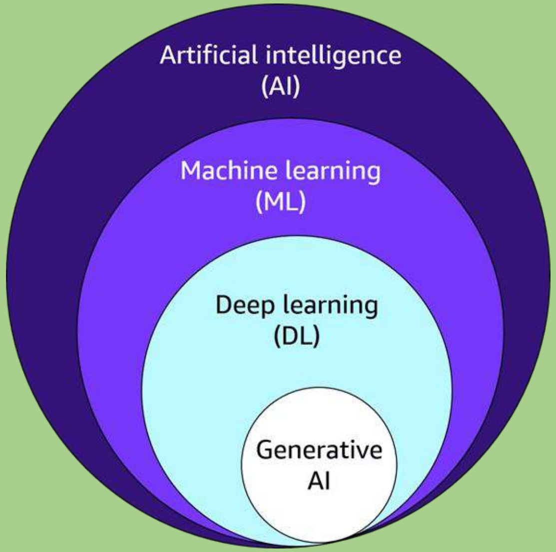
Scenario	Why Use DL	Example
Complex Pattern Recognition	When the task requires understanding intricate patterns from large datasets	Image recognition, speech-to-text, NLP
Handling Unstructured Data	When data is unstructured like images, audio, or text	Video analysis, audio processing
Feature Extraction	When automatic feature learning is needed without manual input	Medical imaging analysis, autonomous driving
Real-time Decision Making	When fast, adaptive decisions are required based on complex inputs	Real-time translation, driver assistance

Generative AI (GenAI)

- **Definition**: Generative AI refers to models that can **create new content** such as text, images, audio, or video based on the patterns it has learned from existing data. These models generate outputs that resemble the data they've been trained on, making them capable of creating entirely new, unique content.
- **How it Works**: Generative AI models, like **GPT-3/4** for text generation or **DALL-E** for image generation, are based on deep learning techniques. These models are often **transformer-based** architectures and are trained on vast amounts of data to learn to generate realistic outputs.
- **Examples**:
  - Text generation (e.g., ChatGPT, GPT-3/4 for generating coherent text)
  - Image generation (e.g., DALL-E for creating realistic images from text descriptions)
  - Music composition (e.g., AI-generated music tracks)
  - Video creation (e.g., AI generating short videos based on input prompts)



Common Use Cases of Gen AI

Gen AI: When To Use ?

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Scenario	Why Use GenAI	Example
Creating New Content	When the task involves generating original text, images, audio, or video	ChatGPT for text generation, DALL·E for images, AI music creation
Enhancing Creativity	When human creativity is augmented or assisted by AI	AI-assisted design, creative writing support
Automating Content Generation	When large-scale content creation is needed with variation	Automated report writing, video generation
Personalized Content	When content needs to be dynamically tailored for individual users	Personalized marketing messages, adaptive learning content

Gen AI: Common Use Cases

Image	Text	Audio	Video	3D/Design	Others
Image Generation	Content Generation	Audio Generation	Video Generation	3D Animation	Sports Commentary
Image Editing	Content Marketing	Speech Synthesis	Face Cloning	3D Asset Generation	NFT Development
Image Enhancement	Content Moderation	Voice Cloning	Video Enhancement	3D Model Generation	Campaign Creation
Image Translation	Language Translation	Audio Denoising	Video Processing	Procedural Generation	Programmatic Advertising
	Text Summarization	Speech Recognition	Video Summarization	Video Game Design	Virtual Content Generation
				3D Scene Generation	